



Advanced Dialogflow Chatbot: Loan + Investment Assistant

AI-Based Financial Guidance using Dialogflow ES

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Bridging the Financial Knowledge Gap



The Problem

Most users struggle to understand loan eligibility criteria, EMI calculations, and investment planning – leading to poor financial decisions.

Our Solution

An AI-powered finance chatbot using Dialogflow ES that delivers instant, conversational guidance on loans, EMIs, and investments (SIP).

Core Objectives

- Automate finance-related responses using NLP
- Improve user interaction through conversational AI
- Demonstrate practical chatbot and intent-based technologies

AI & NLP in Finance

What is AI in Finance?

Artificial Intelligence (AI) refers to the use of smart algorithms and machine learning models to automate financial decision-making, analyze large datasets, and improve efficiency in banking and investment systems.

What is NLP (Natural Language Processing)?

Natural Language Processing (NLP) is a branch of AI that enables machines to understand, interpret, and respond to human language in a conversational manner.

Role of AI & NLP in Finance

- Automates customer interactions through chatbots
- Provides real-time financial advice
- Enhances fraud detection and risk analysis
- Improves decision-making using data-driven insights
- Enables personalized investment recommendations

Applications in Your Project

- Chatbot understands user queries like: "Can I get a loan?" and "Suggest low-risk investment"
- NLP processes user intent using Dialogflow
- AI provides accurate responses based on financial logic
- Reduces human effort and increases response speed

Real-World Examples

- Banking chatbots for customer support
- AI-based loan approval systems
- Robo-advisors for investment planning
- Voice assistants for financial queries

Why It Matters

Faster

Smarter

More accessible

User-friendly

MODULE 1

Loan Eligibility Intent

The screenshot shows the Dialogflow console for the 'Loan_Eligibility_Advanced' intent. The left sidebar contains navigation links: Dialogflow Essentials, Finance_Assistant_Adva..., Intents (selected), Entities, Knowledge, Fulfillment, Integrations, Training, Validation, History, Analytics, Prebuilt Agents, and Small Talk. The main content area is titled 'Loan_Eligibility_Advanced' and includes a 'SAVE' button. It has sections for 'Contexts', 'Events', 'Training phrases', and 'Action and parameters'. A yellow warning box states: 'Template phrases are deprecated and will be ignored in training time. More details here.' Below this, a training phrase is listed: 'I earn 50000 can I get loan. My salary is 70000 what loan can I get. Loan eligibility for 60000 income. Can I get loan of 20 lakh. I earn 80000 and want home loan. My monthly income is 45000'. The 'Action and parameters' section is currently empty. On the right, a preview window shows a simulated conversation with the agent, including a recommendation for a loan and a calculation of EMI.

Intent: `Loan_Eligibility_Advanced`

Assesses loan eligibility based on user-provided monthly income and financial profile.

Sample Training Phrases

- "I earn 50000 can I get a loan?"
- "Loan eligibility for 60000 income"
- "I earn 80000 and want a home loan"

Parameters

- `income` → `@sys.number` (Required)
- `loan_amount` → `@sys.number` (Optional)

Eligibility Factors

Monthly income · Credit score · Existing EMI's ·
Employment type · Repayment history
Age Eligibility

MODULE 2

EMI Intent

Intent: `EMI_Calculation`

Helps users understand possible EMI ranges and loan affordability based on their requirements.

Bot Calculates EMI

Use this EMI Formula :
$$EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$$

Parameter

`loan_amount` → `@sys.number`

The screenshot displays the Dialogflow console interface for configuring an intent named "EMI_Calculation". The left sidebar shows the navigation menu with options like Intents, Entities, Knowledge, Fulfillment, Integrations, Training, Validation, History, Analytics, Prebuilt Agents, Small Talk, and Docs. The main area shows the intent configuration for "EMI_Calculation". A warning message states: "Template phrases are deprecated and will be ignored in training time. More details here." Below this, there is a section for "Add user expression" with a list of training phrases: "Calculate EMI for 1000000 loan", "EMI for 500000", "What will be EMI for 20 lakh loan", "Calculate monthly installment", and "I want EMI estimate". The "Action and parameters" section is expanded, showing a table of parameters:

REQUIRED	PARAMETER NAME	ENTITY	VALUE	IS LIST	PROMPTS
☒	loan_amount	@sys.number	loan_amount	☒	Please enter th...
☐	loan_amount1	@sys.number	loan_amount1	☐	—
☐	Enter name	Enter entity	Enter value	☐	—

Below the table, there is a "+ New parameter" link. On the right side of the console, a chat window is visible with the agent "Finance_Assistant_Advanced". The chat history shows a user asking for investment suggestions, and the agent responding with recommendations for low and high risk, including a list of recommended plans (Equity Mutual Funds, Direct Stocks, ETFs, Expected Returns, SIP recommendations) and a prompt to calculate EMI for 1000000 loan.

MODULE 3

Investment Intent

Intent: `Investment_Advanced` | Custom Entity: `risk_level`

Conservative Preference

Fixed Deposits, Bonds, Debt
Mutual Funds

Balanced Preference

Balanced Funds, Index Funds,
Hybrid Schemes

Growth Preference

Stocks, Equity Mutual Funds, Derivatives, SIP (Systematic Investment Plan)

Sample Phrases: "I want safe investment" · "I can take high risk" ·
"Investment for medium risk"

The screenshot displays the Dialogflow console for an agent named 'Investment_SIP'. The left sidebar contains a navigation menu with options: Intents, Entities, Knowledge, Fulfillment, Integrations, Training, Validation, History, Analytics, Prebuilt Agents, Small Talk, and Docs. The main workspace is divided into two primary sections: 'Action and parameters' and 'Responses'. The 'Intents' tab is selected, showing a table for defining parameters with columns for 'REQUIRED', 'PARAMETER NAME', 'ENTITY', 'VALUE', and 'INPUT'. Below the table, there is a 'New parameter' button. The 'Responses' section includes a 'Text Response' area with a list of responses and an 'ADD RESPONSES' button. On the right side, a chat window titled 'Finance_Assistant_Advanced' shows a simulated conversation. The chat history includes a user query about loan applications, an agent response suggesting a loan, and a user query about investment preferences. The agent responds with a recommendation for a high-risk plan and provides details about equity mutual funds and ETFs. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 11:08 AM on 5/6/2023.

LOGIC FOR EMI CALCULATION

Bot Calculates EMI

Use this EMI Formula : $EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$

Where:

- P = Loan Amount
- * R = Monthly Interest Rate
- * N = Number of Monthly Instalments
- Example Calculation
- If: * Loan = ₹5,00,000
- * Interest = 10%
- * Tenure = 5 years
- Then:
- * Monthly EMI ≈ ₹10,624
- * Total Payment ≈ ₹6.37 Lakhs

LOGIC FOR SIP CALCULATION

Bot Calculates SIP Return

SIP Formula:

$$M = P \times \frac{\left(1 + \frac{r}{n}\right)^{nt} - 1}{\frac{r}{n}} \times \left(1 + \frac{r}{n}\right)$$

Where:

P = Monthly Investment

* r = Annual Return

* t = Time in years

* n = 12 months

Example Output

If: * SIP = ₹5,000/month

* Duration = 10 years

* Return = 12%

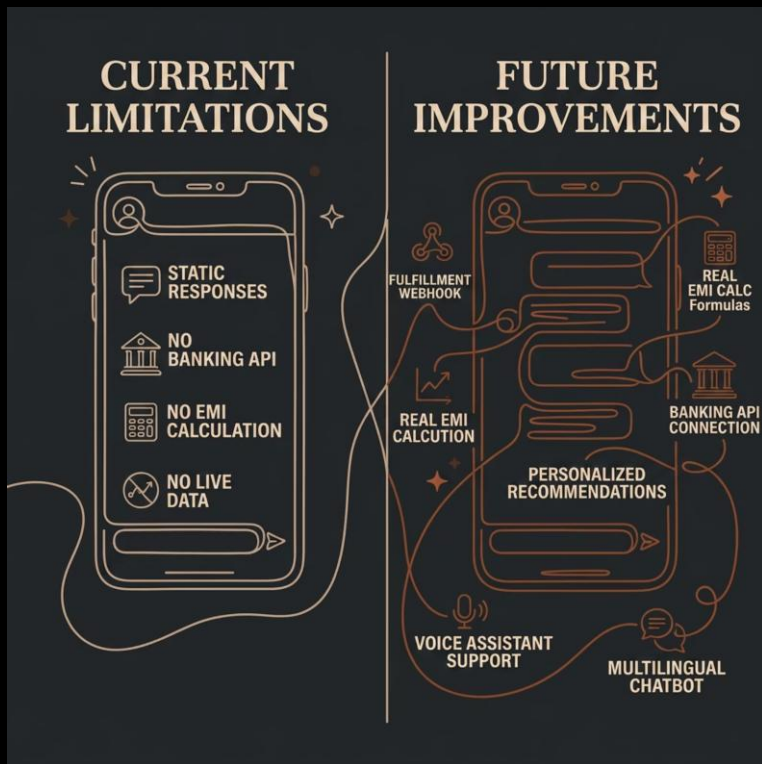
Then:

* Total Invested = ₹6 Lakhs

* Estimated Returns ≈ ₹5.6 Lakhs

* Final Amount ≈ ₹11.6 Lakhs

Limitations & Future Improvements



What We're Building Towards

The current version demonstrates core NLP capabilities. Future iterations will connect to live financial APIs, enable real-time calculations via webhooks, and support voice and multilingual interactions for broader accessibility.



Next milestone: Integrating a Fulfillment Webhook to enable dynamic, real-time financial computations.

Conclusion

Key Achievements

1

Built an intelligent chatbot using Dialogflow ES

2

Applied NLP to understand user queries accurately

3

Automated financial guidance with real-time responses

4

Improved user experience through simple conversation flow

Business Impact

Reduces dependency on human financial advisors

Saves time and operational cost for financial institutions

Provides instant, 24/7 financial assistance

Makes financial knowledge accessible to a wider audience

Learning Outcomes

→ Practical understanding of AI and NLP in finance

→ Hands-on experience with chatbot development

→ Improved knowledge of financial decision-making concepts

→ Integration of technology with real-world applications

- [Tripti GitHub](#)
- [saransh github](#)
- [khushboo github](#)
- https://github.com/nakshatra2654mba25fin-sketch/ai-finance-expense-automation-system/blob/main/Lab_1_2_Bias_Detection_Loan_Approval.ipynb